

Development of an automated mobile grain size mapping of a mixed sediment beach

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Abstract

Grain sizes influence beach morphologies, processes, and states (e.g., dissipative, reflective and intermediate), however previous studies analysing beach grain size are often spatially and temporarily sparse. In this study, an automated mobile grain size estimation method using digital photographs (mobile digital grain size, MDGS) was developed and tested at a mixed sediment beach consisting of sand and gravels. The data collection system consisted of a downward-looking camera, Real Time Kinematic Global Navigation Satellite System, and a camera-triggering single-board computer, attached to an all-terrain vehicle. Digital photographs were collected monthly from January 2021 to March 2022 over a section spanning ~700 m along-shore and processed automatically using an existing (non-mobile) DGS estimation method. The MDGS results were generally consistent with manually estimated grain sizes ($n = 962$, $r^2 = 0.74$, $RMSE = 5.9$ mm) and UAV orthomosaic imagery, although the MDGS sometimes overestimated manually estimated grain sizes particularly for photographs with relatively small sediments. Modelled nearshore waves and subaerial beach survey data from mobile terrestrial LiDAR were used to explore using MDGS in morphological studies. Seasonal grain size trends, observed relationships between nearshore wave height and spatially averaged beach grain size, and the general increasing trends of grain sizes with both beach elevation and local roughness/slope were consistent with previous studies, highlighting the use of MDGS for morpho-sedimentary mixed sediment beach research. The MDGS provides a unique method to conduct spatially and temporarily high-resolution mapping of gravel size sediments over large areas ($>10^2$ m alongshore). Potential improvements of the MDGS include additional low elevation cameras for improved fine (e.g., sand) grain size estimates, decreased time between photo triggering, and automated removal of erroneous photographs containing surface irregularities (e.g., tire tracks) and non-sediment features (e.g., seaweed). Overall, the MDGS provides improved surface grain size mapping and investigations of complex grain size-morphology relationships of mixed sediment beaches.

KEYWORDS

coastal, digital photograph, grain size, mixed sediment beach, morpho-sedimentary

1 | INTRODUCTION

Grain size is a fundamental property of beaches. It is used for simple first-order classification (e.g., sandy beach, gravel beach, and mixed sediment beach) and also greatly influences beach morphologies (e.g., Carter & Orford, 1993; Nolan et al., 1999), processes (e.g., Buscombe & Masselink, 2006; Holland & Elmore, 2008; Kirk, 1980), and states (i.e., dissipative, intermediate, reflective) (e.g., Masselink & Short, 1993; Wright & Short, 1984). For example, coarser sediments generally form steeper beach slopes (e.g., Bascom, 1951; McLean & Kirk, 1969). Coarse sediments are also generally more stable to wave attack and have been used for 'nature-based' shoreline protection strategies (e.g., Allan & Komar, 2004). Improved spatial and temporal resolution of variable grain sizes are needed to better simulate sediment transport (Buscombe & Masselink, 2009) and morphological changes of mixed sediment beaches (Curtiss et al., 2009).

Because of the relatively time-consuming and laborious nature of traditional grain size analysis techniques such as sieve analysis, previous beach grain size studies were often spatially and temporarily limited. However, beach grain size can vary considerably both in space and time. For example, on sandy beaches, Prodger et al. (2017) reported grain size coarsening and enhanced sorting with increasing seaward distance across the intertidal zone particularly during winter months. For mixed sediment beaches, Allan et al. (2006), Matsumoto et al. (2020b) and Casamayor et al. (2022) found strong seasonality in coarse sediment exposure and corresponding beach profile changes at composite beaches consisting of a sandy foreshore and coarse sediment backshore (Jennings & Shulmeister, 2002). Matsumoto et al. (2020a) also observed the dynamic nature of composite mega cusps, composed of sandy horns and coarse sediment bays, where horn and bay locations completely exchanged on monthly time scales.

Recent advances in automated grain size analysis techniques include remote sensing methods. Several studies used LiDAR and hyperspectral cameras attached to mobile survey platforms such as airplanes and ground-based vehicles to automatically map and classify generalized regions of different (often coarse) grain sizes over spatially large areas (e.g., Deronde et al., 2008; Gómez et al., 2022; Matsumoto & Young, 2018). Other approaches used digital photographs to automatically identify and calculate individual grain sizes (i.e., object-based image analysis, Adams, 1979; Graham et al., 2005; Carboneau et al., 2018), or calculate average grain sizes for each photograph using statistical analysis (e.g., Barnard et al., 2007; Buscombe et al., 2010; Carboneau et al., 2004; Rubin, 2004). The average grain size estimation approach has been commonly applied to beaches with variable grain sizes (e.g., Guest & Hay, 2021; McFall et al., 2020; Pentney & Dickson, 2012; Prodger et al., 2017; Warrick et al., 2009), although these previous studies generally collected digital photographs on foot making them spatially limited.

In this study, an automated mobile digital grain size estimation method using a statistical approach applied to digital photographs (MDGS) was developed and tested on a mixed sediment beach consisting of sand and coarse sediments, including pebbles (4–64 mm) and cobbles (64–256 mm) (denoted as gravels in this study after Wentworth, 1922). The MDGS was validated against manual grain size estimations using digital photographs and an unmanned aerial vehicle (UAV) imagery. The MDGS results were examined in relation

to seasonal differences, nearshore waves and beach morphologies to explore using MDGS in morph-sedimentary beach studies. The discussion explores advantages and limitations of the MDGS compared with existing methods and identifies refinements for improved MDGS mapping.

2 | MATERIAL AND METHOD

2.1 | Study site

The study site spans a ~700-m alongshore section of Torrey Pines State Beach in southern California that consists of sand ($D_{50} = 0.23$ mm; Ludka et al., 2015) and gravels ($D_{50} = 52.7$ mm, longest axis; Matsumoto et al., 2020a). The beach is backed by riprap in the north and sea cliffs in the central and southern sections. Sand generally accretes in summer and erodes in winter owing to the seasonal fluctuations of wave energy (e.g., Aubrey, 1979; Ludka et al., 2015). Surface gravel exposure is also seasonal with more gravels exposed during winter months (Matsumoto et al., 2020b; Matsumoto & Young, 2022). Gravels commonly accumulate near the back beach, creating steep gravel berms fronted by a low-slope sandy foreshore, which Jennings and Shulmeister (2002) classify as a composite beach. However, gravel exposure varies both in space and time with the internal sediment of upper foreshore often ranging from moderately to fully mixed sand and gravel (down to <1–2 m deep), which Jennings and Shulmeister (2002) classify as a mixed sand and gravel beach. The tidal range (mean higher high water–mean lower low water) is 1.6 m, with mean sea level (MSL), mean high water (MHW), and highest astronomical tides of 0.8, 1.3 and 2.1 m (North American Vertical Datum 1988, navd88), respectively. A lagoon mouth exists at the northern end of the study area, where sediment dredging takes place annually (generally in spring) to move sediment from the lagoon mouth to adjacent beaches.

2.2 | MDGS

2.2.1 | Mobile photo collection system

A downward looking Nikon P900 digital camera (sensor size: 6.17×4.55 mm, image size: 4608×3456 pixels) was attached to an all-terrain vehicle (ATV) about 1 m above the ground. The ATV drove at around 8 km/h, while an Arduino single-board computer triggered the camera every 3 s (Figure 1). A focal length of 24 mm was selected based on qualitative preliminary testing resulting in ground sampling distance (GSD) of 0.05 mm per pixel and 26×19 cm photograph ground footprint with a maximum measurable grain size of ~19 cm. The image distortion from the camera lens was found to be negligible. A plastic plate with a hole was used to hold the lens at a fixed elevation relative to the camera elevation and prevent lens movement from ATV vibration. A shutter speed of 1/3200 was selected to minimize blur from mobile photo collection, while an embedded flash and the ISO sensitivity of either 1600 (sunny) or 3200 (cloudy) was used to maintain similar lighting conditions.

The system included a Real Time Kinematic Global Navigation Satellite System (RTK-GNSS) with a centimetre level accuracy (Figure 1). Photo locations were determined using the photo

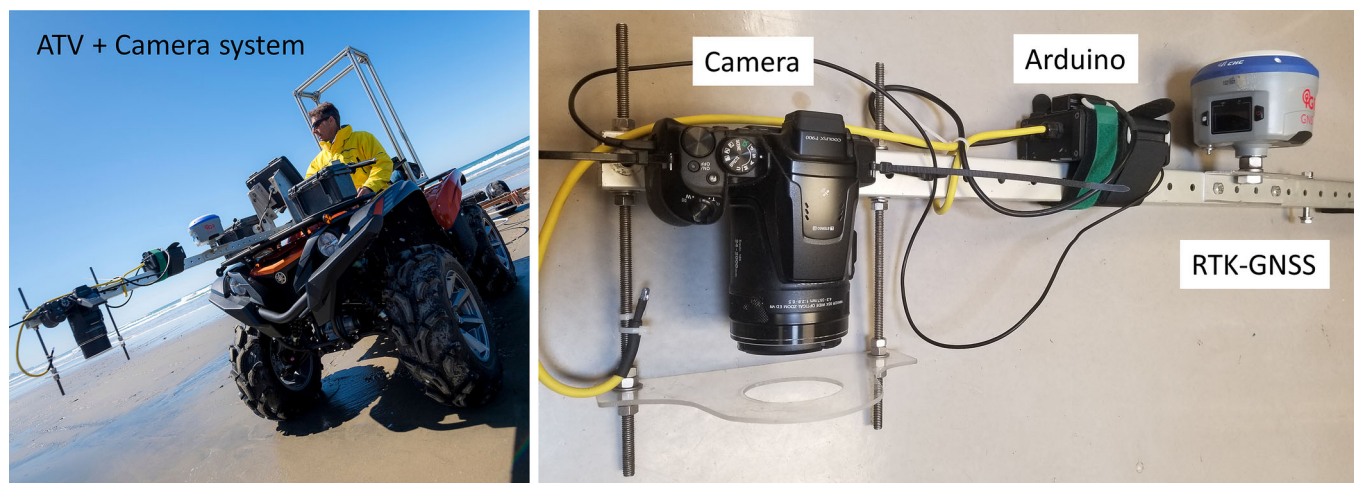


FIGURE 1 Images showing the mobile photo collection system including all-terrain vehicle (ATV), Nikon P900 digital camera, Arduino single-board computer, and Real Time Kinematic Global Navigation Satellite System (RTK-GNSS). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/esp.5655)]

acquisition time from the on-board camera GPS and corresponding RTK-GNSS positions. Photo acquisition information was limited to 1-s sampling point spacing generating a maximum of 2.2 m ($8 \text{ km/h} \times 1 \text{ s}$) horizontal position errors.

The mobile photo collection occurred monthly from January 2021 to March 2022 during low tides. The ATV drove the beach both alongshore and cross-shore with $\sim 2\text{--}3 \text{ m}$ spacing between passes, taking about 2–3 h to cover the entire study section. Photo collection was not conducted in June and September 2021, and January 2022 because of a camera triggering problem. The camera triggering problem also occurred during a few other surveys hindering continuous mobile photo collection occasionally. In total, 24 415 digital photographs were collected (775–2443 photographs per survey) with an overall mean ground sampling density of 0.07 photograph per square meter (ranging 0.04–0.14 per survey). The photo sampling distance was generally insufficient for generating seamless mosaic maps.

2.2.2 | DGS analysis

This study used DGS analysis method and associated open-source MATLAB program developed by Buscombe et al. (2010) that applies Fourier analysis to space-series transects of greyscale pixel intensities in the digital photographs. For each digital photograph, DGS automatically estimates (image processing time of a few seconds) the mean and percentiles (e.g., D_{50}) of grain sizes and the grain size distribution using log-hyperbolic distribution density function fitting (e.g., Bader, 1970). The method does not require calibration, but does require GSD (mm per pixel) that converts estimated grain sizes to real-world size. The GSD depends on the distance of the camera to the ground (and a lens zoom setting) and can vary owing to ATV/camera vibrations. During initial testing, manual counting of the number of pixels along a known ground distance confirmed GSD varied little ($\pm 2\%$) even during ATV operation. This study also tested upgraded DGS analysis methods (Buscombe, 2013, 2020) but found the upgraded methods did not significantly improve the results.

Some results were automatically omitted because of potential errors. These included results from blurry photographs, and where meso-scale (e.g., $10^1\text{--}10^2 \text{ cm}$) topographic irregularities

(e.g., beach depressions, Figure 2) can alter the camera elevation from the ground and associated GSD. Blurriness was quantified as the variance of Laplacian method using an open-source Python program (github.com/WillBrennan/BlurDetection2) (Pech-Pacheco et al., 2000). Values of 200–500 blurriness were tested, and a blurriness threshold of 200 minimized the difference between MDGS and manual grain size estimations and used in this study. To identify photographs with the meso-scale topographic irregularities, a best fit plane was estimated for each photograph using the ground elevations (from LiDAR data, described below) below the camera and ATV wheels (based on GPS locations) at the timing of photo-collection (Figure 2). This study omitted results when at least one of the ground elevations deviated $>10 \text{ cm}$ from the best fit plane. Results were also omitted when the log-hyperbolic density function fitting failed possibly because of a limited number of grains (e.g., <1000) contained in the digital photographs (see Buscombe et al., 2010, for more details). In total, 18 126 photographs were omitted and 6289 were examined providing a mean survey sample density of 0.02 (ranging 0.002–0.040) photographs per square meter.

2.3 | Manual grain size estimation

To examine MDGS results, grain sizes were manually estimated using 962 randomly selected digital photographs. These photographs were first manually classified into three classes: sand-only (461), gravel-only (212), and sand–gravel mix (289) (Figure 3a–c). For gravel-only photographs, up to 100 gravel grains were randomly selected in each photograph, and intermediate axes were measured (red lines, Figure 3a) to calculate mean and median gravel grain sizes (after Barnard et al., 2007; Buscombe et al., 2010). For sand-only photographs, individual grains were identified using a MATLAB image segmenter tool, and intermediate axes were estimated using a MATLAB image region analyser tool. Sometimes, the method did not work for sand-only photographs with relatively large blurriness. This study assumed sand grain sizes were relatively less variable compared with gravel grain sizes and assigned a mean sand grain size of 1.1 mm to all sand-only photographs based on manual estimations of 10 sand-only photographs that ranged 0.95–1.23 mm with a mean of 1.1 mm. For

FIGURE 2 Example two-dimensional sketch of meso-scale topographic irregularity, and its relation to the elevations of the all-terrain vehicle (ATV)-attached camera and ATV wheels. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

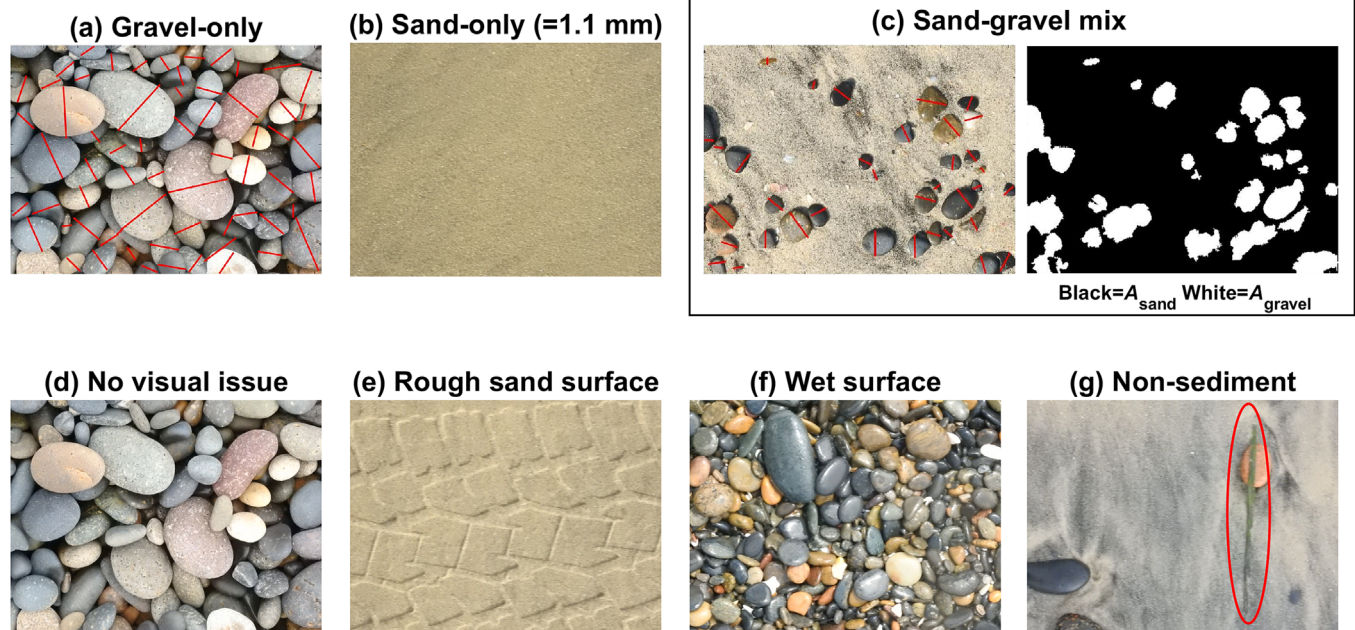
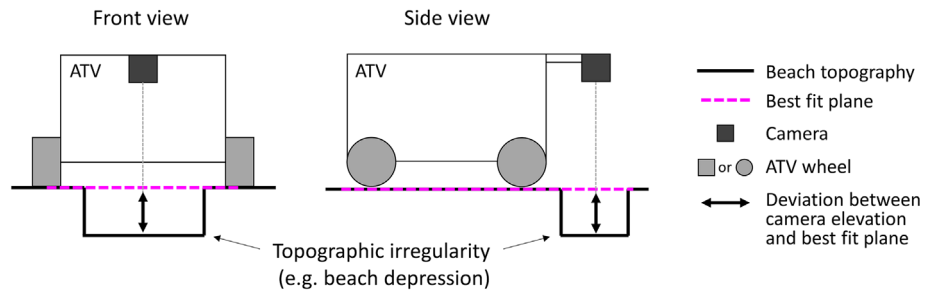


FIGURE 3 Example photographs of (a) gravel-only, (b) sand-only, and (c) sand-gravel mix classes, and sub-classes of (d) no visual issue, (e) rough sand surface, (f) wet surface, and (g) non-sediment. Red lines in (a, c) show measured intermediate axes, and a red circle in (g) shows the location of a non-sediment feature (i.e., seaweed). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

sand-gravel mix photographs, a mean gravel grain size (GS_{gravel}) was manually estimated following the steps for gravel-only photographs, and the overall mean grain size of a sand-gravel mix photograph ($GS_{\text{sand-gravel}}$) was estimated using:

$$GS_{\text{sand-gravel}} = 1.1 \times A_{\text{sand}} + GS_{\text{gravel}} \times A_{\text{gravel}}$$

where A_{sand} and A_{gravel} are relative sand and gravel areas, respectively, and were determined using the MATLAB image segmenter tool (Figure 3c). In total, 23 308 gravel grains were manually measured. To further explore possible error factors, the 962 photographs were manually classified into four sub-classes: (1) no visual issue, (2) micro-scale (e.g., 10^0 – 10^1 cm) sand surface irregularities such as ATV tire tracks (denoted as rough sand surface in this study), (3) wet surface, and (4) non-sediment (e.g., seaweed) (Figure 3d–g).

2.4 | LiDAR and UAV photo surveys

Mobile terrestrial LiDAR surveys were conducted concurrently with the mobile photo collection. The LiDAR system consisted of RIEGL VZ-2000 laser scanner, a RTK-GNSS and an inertial measurement unit (IMU), providing a survey accuracy of a few centimetres. LiDAR point

clouds were ground-filtered (Olsen et al., 2020) and manually edited to remove erroneous points such as waves, birds and people. Point clouds were processed with a direct georeferencing solution (navd88) to generate elevation, roughness and slope rasters at 0.2-m GSD for morphological analysis. The roughness raster was calculated as the deviation of the individual point elevations from the mean elevation, while the slope raster was calculated as an average slope considering mean elevations of the eight surrounding raster pixels.

UAV high resolution photo surveys (with ~80% front and side overlap) were also conducted concurrently with the ground-based mobile photo collection using a DJI Phantom-4 RTK with a 20-MP camera. Eight ground control points were surveyed using RTK-GNSS. Orthomosaic imageries were generated using Pix4Dmapper and used to visually examine the MDGS estimations.

2.5 | Nearshore waves

A buoy-driven, regional wave model was used to estimate hourly significant wave height (H_s) in 10 m water depth offshore location of the study site (O'Reilly et al., 2016). The model includes the effects of complex offshore bathymetry and varying beach orientation on wave exposure.

3 | RESULTS

3.1 | MDGS versus manual grain size estimation

MDGS results were correlated with manual mean grain size estimations with $r^2 = 0.74$ and root mean square error (RMSE) = 5.9 mm

(Figure 4). In addition, the probability distribution functions of sand-only, sand-gravel mix, and gravel-only results were qualitatively similar between the MDGS and manual estimations. However, the MDGS results generally overestimated manual results (mean of +3.5 mm for sand-only, +7.6 mm for sand-gravel mix and +6.5 mm for gravel-only), and MDGS exhibited relatively large scatter compared with

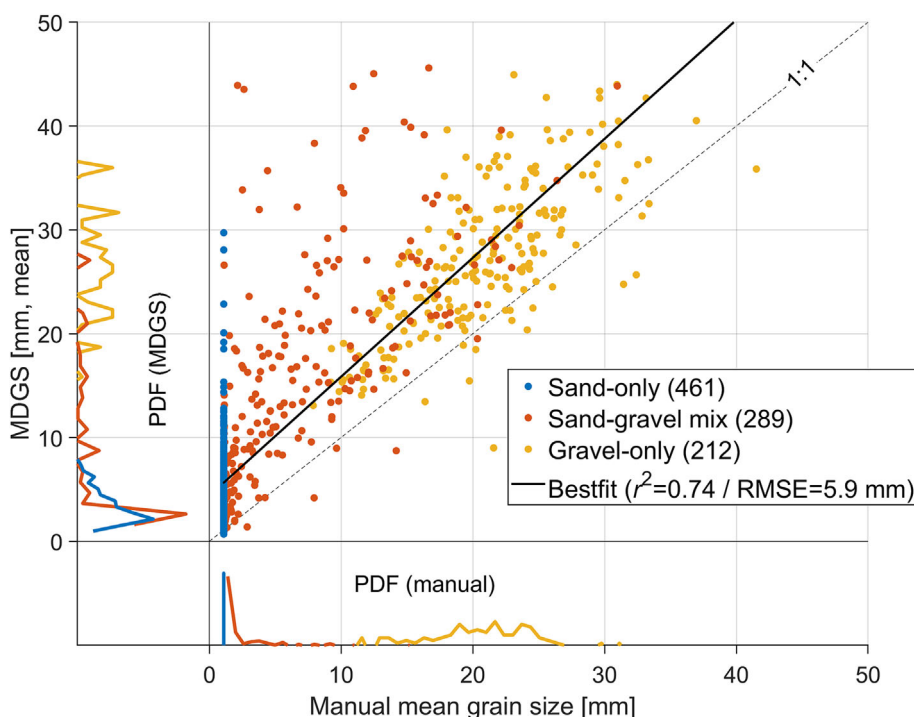


FIGURE 4 Comparison of mobile digital grain size (MDGS) and manual estimations for each photograph, and associated probability distribution functions for sand-only, sand-gravel mix, and gravel-only classes. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/esp.5655)]

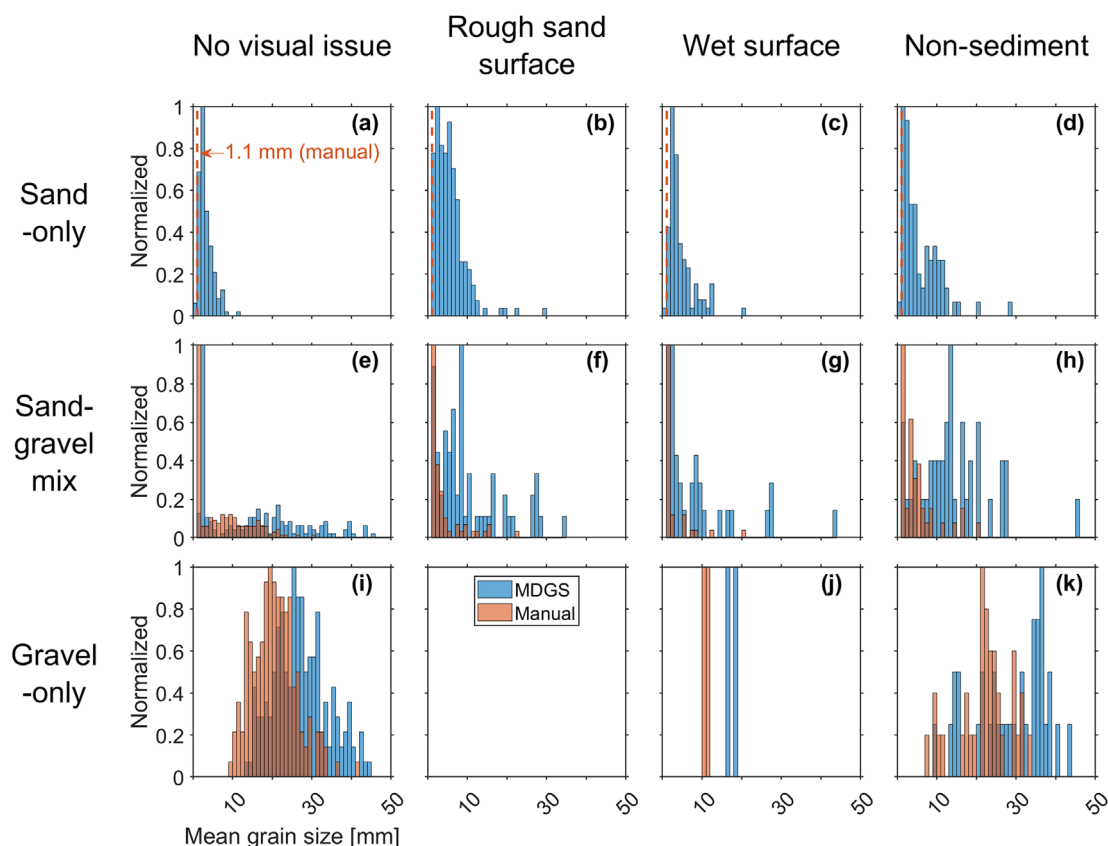


FIGURE 5 Comparison of normalized (by the maximum count) histograms of mobile digital grain size (MDGS) (blue) and manual grain size (orange) results for (a–d) sand-only, (e–h) sand-gravel mix and (i–k) gravel-only photographs for each sub-class of ‘no visual issue’, ‘rough sand surface’, ‘wet surface’ and ‘non-sediment’. The vertical dashed orange lines in (a–d) show the manual results. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/esp.5655)]

manual estimation particularly for sand-only and sand-gravel mix results (similar to Buscombe, 2020). For example, mean grain sizes from the MDGS using sand-only and sand-gravel mix photographs were up to 15 and 18 times higher than manual estimations, respectively. In contrast, the differences between the MDGS and manual estimations for gravel-only photographs were relatively small with MDGS mean grain sizes up to about two times higher compared with manual estimates. Median gravel-only results were similar to mean grain size results, with +8.4 mm mean overestimation of MDGS compared with manual estimations (not shown).

For 'no visual issue' sand-only photographs, the normalized distribution of the MDGS results was narrow and peaked around 2–3 mm (Figure 5a), whereas distributions for other sub-classes were broader, particularly for sand-only photographs with 'rough sand surface' and 'non-sediment' (Figure 5b–d). For sand-gravel mix photographs, the manual results peaked at around 1–2 mm regardless of the sub-classes. In contrast, only the MDGS results with 'no visual issue' and 'wet surface' sub-classes peaked near the 1–3 mm (Figure 5e–h). The distributions were similar between the MDGS and manual estimations

for gravel-only photographs with 'no visual issue' and 'non-sediment', although the MDGS generally overestimated manual results (Figure 5i,k). There were very few gravel-only photographs with 'wet surface' ($n = 2$) (Figure 5j).

3.2 | Comparison with UAV orthomosaic imagery

The spatial (surface) grain size distribution from the MDGS was qualitatively consistent with beach sediments observed from the UAV orthomosaic imagery (Figure 6). For example, coarse sediments with 20–50 mm mean grain sizes were frequently found at around 200–300 and 500–600 m alongshore locations in the MDGS results (green circles, Figure 6b), consistent with the orthomosaic imagery (green circles, Figure 6a). Similarly, fine sediments were frequently found using both methods at around 350–500 m alongshore locations (red circles, Figure 6a,b). However, the spatial resolution of the MDGS was not uniform and included some spatial data gaps (i.e., white areas, Figure 6b).

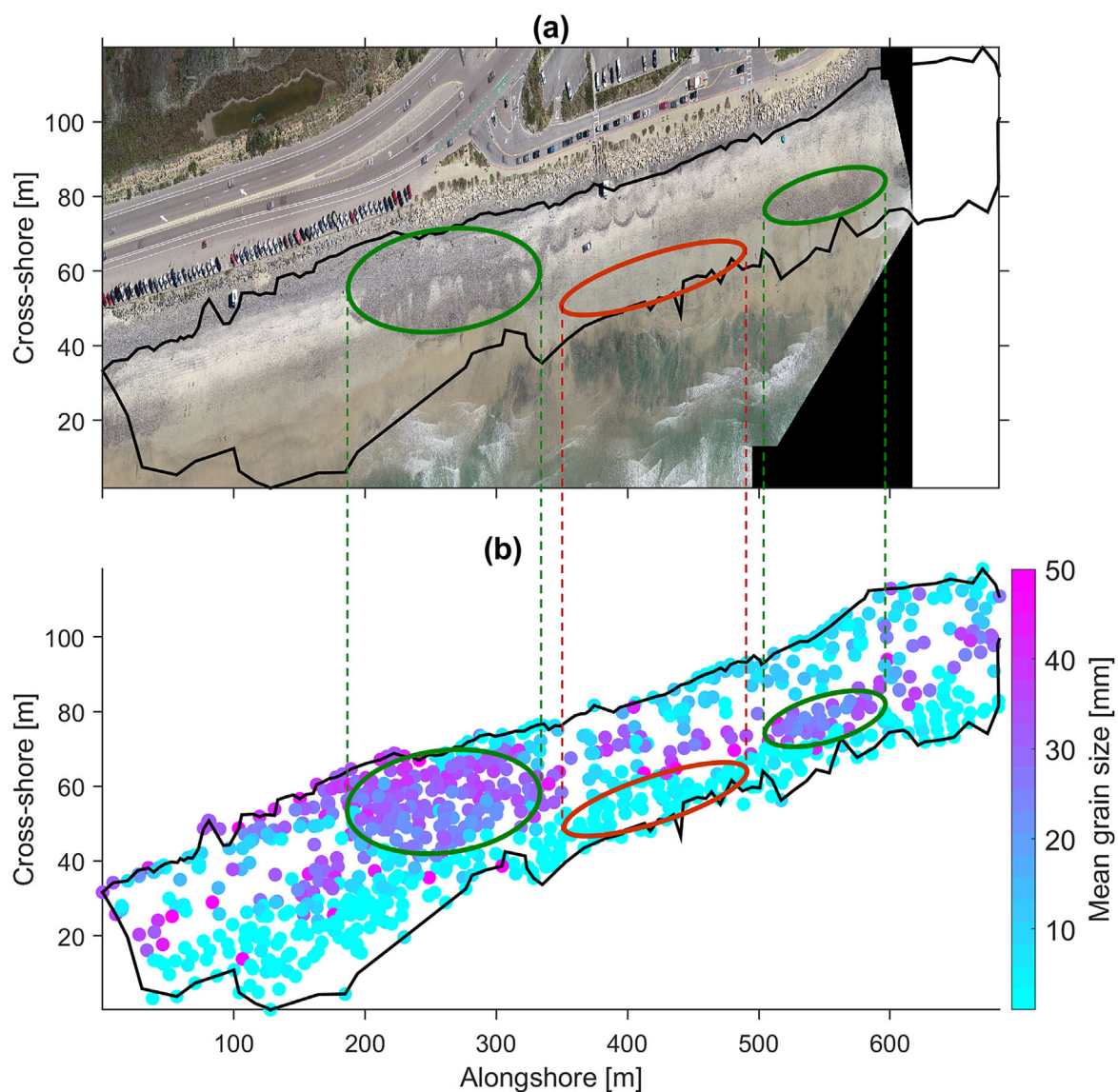


FIGURE 6 Spatial distribution of variable grains from the March 2022 (a) all-terrain vehicle (UAV) orthomosaic imagery and (b) mobile digital grain size (MDGS). Green and red circles show the example locations of coarse and fine sediments, respectively. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3.3 | Seasonal grain size trends and comparison with waves and beach topography

The spatial distributions of fine/coarse sediment (e.g., sand/gravel) surface exposures determined using the MDGS varied seasonally (Figure 7). For example, the beach was generally sandy in summer and fall with coarse sediments found only at the back beach (Figure 7a,b). In early-mid winter, coarse sediments occurred on the upper beach-face but was often spatially intermittently (e.g., Figure 7c) resulting in relatively spatially scattered upper beach coarse sediment exposures in December–February (Figure 7d). The beach was often narrower and composed of more coarse-sediment in late winter (e.g., Figure 7e) with MDGS coarse sediments frequently observed almost across the entire study section in March–April (Figure 7f).

Spatially averaged beach grain sizes generally increased with nearshore H_s (Figure 8). For example, both the monthly H_s and spatially averaged beach grain sizes were the lowest in July and August 2021 and started to increase in October 2021. Monthly H_s was higher between January and March 2021 (with the monthly peak of ~ 1.5 m in January 2021) when the beach grain sizes were relatively large.

However, the largest (>15 mm) beach grain sizes were observed late in winter between March and May 2021 (similar to Figure 7e,f). The observed summer-low and winter-high beach grain sizes are likely because smaller sand grains are often eroded from the subaerial beach during winter while many relatively large grain size sediments remain on the beach (e.g., Evert et al., 2002).

Beach elevation, local roughness and local slope increased with the MDGS mean grain size estimates (Figure 9). For example, MDGS grain size of 0–20 mm was frequently found at elevations <2.5 m (navd88), whereas the median elevation increased with grain sizes for large grains (20–50 mm, Figure 9a). Local roughness and slope generally increased for mean grain sizes ranging 0–20 mm, although the trends were relatively less clear for larger mean grain sizes ranging 20–50 mm (Figure 9b,c). There were more grain size outliers for the roughness and slope compared with elevation.

Cross-shore profiles extracted from the March 2022 LiDAR survey and MDGS grain sizes (Figure 6b) show a complex relationship between local slope and (surface) grain sizes (Figure 10). For example, the slope of Profile-1 and Profile-2 generally increased with grain size up to around 30 m in cross-shore position where the berm crests

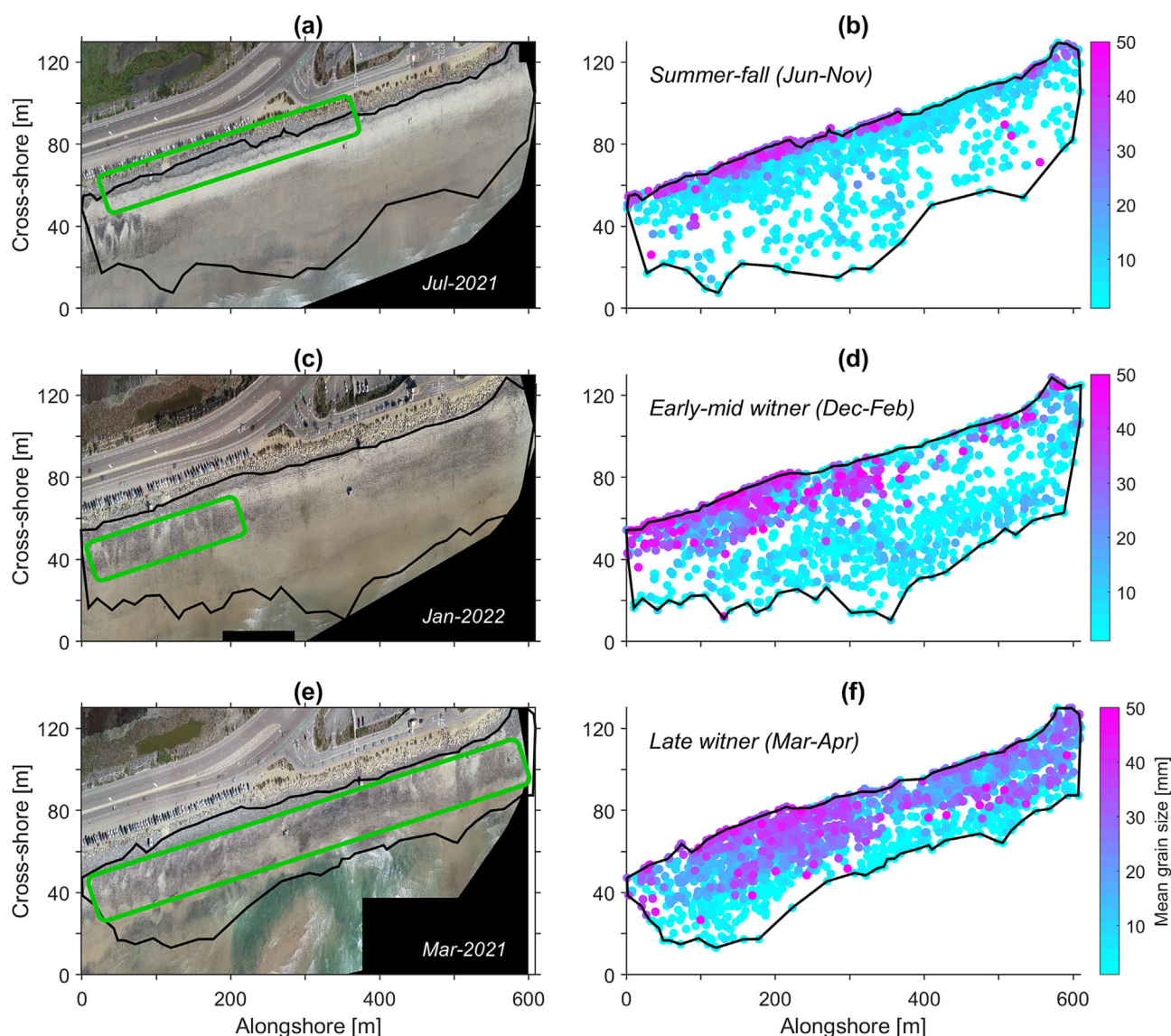


FIGURE 7 Example orthomosaic all-terrain vehicle (UAV) imagery and combined mobile digital grain size (MDGS) results from (a, b) summer-fall, (c, d) early-mid winter, and (e, f) late winter months. Green boxes (in a, c, e) show the example locations of coarse sediments. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/esp.5655)]

FIGURE 8 Time series of (a) hourly and monthly nearshore H_s , and (b) spatially averaged beach grain size. February 2021 mobile digital grain size (MDGS) result is omitted owing to a small number of results ($n < 100$). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

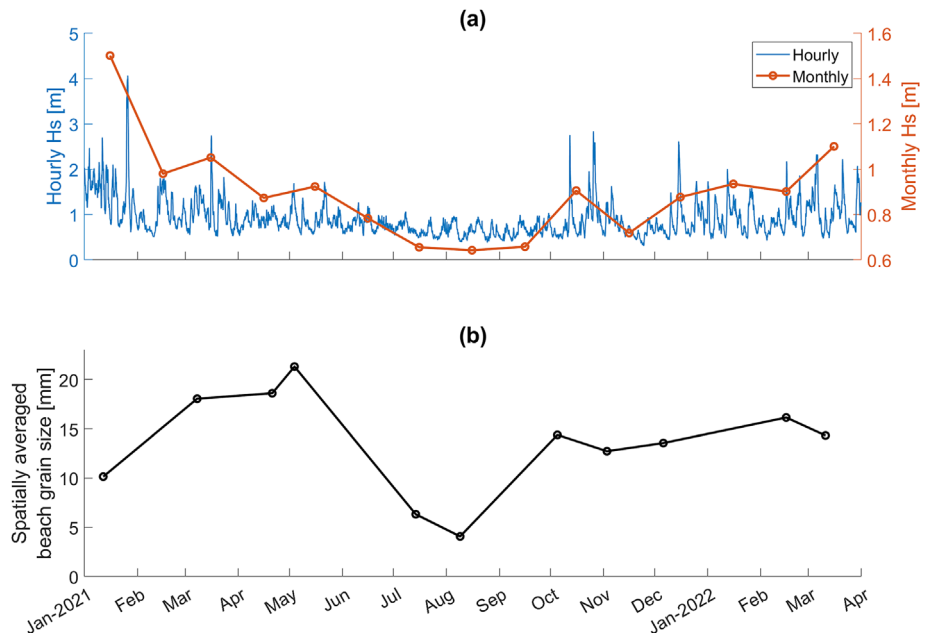
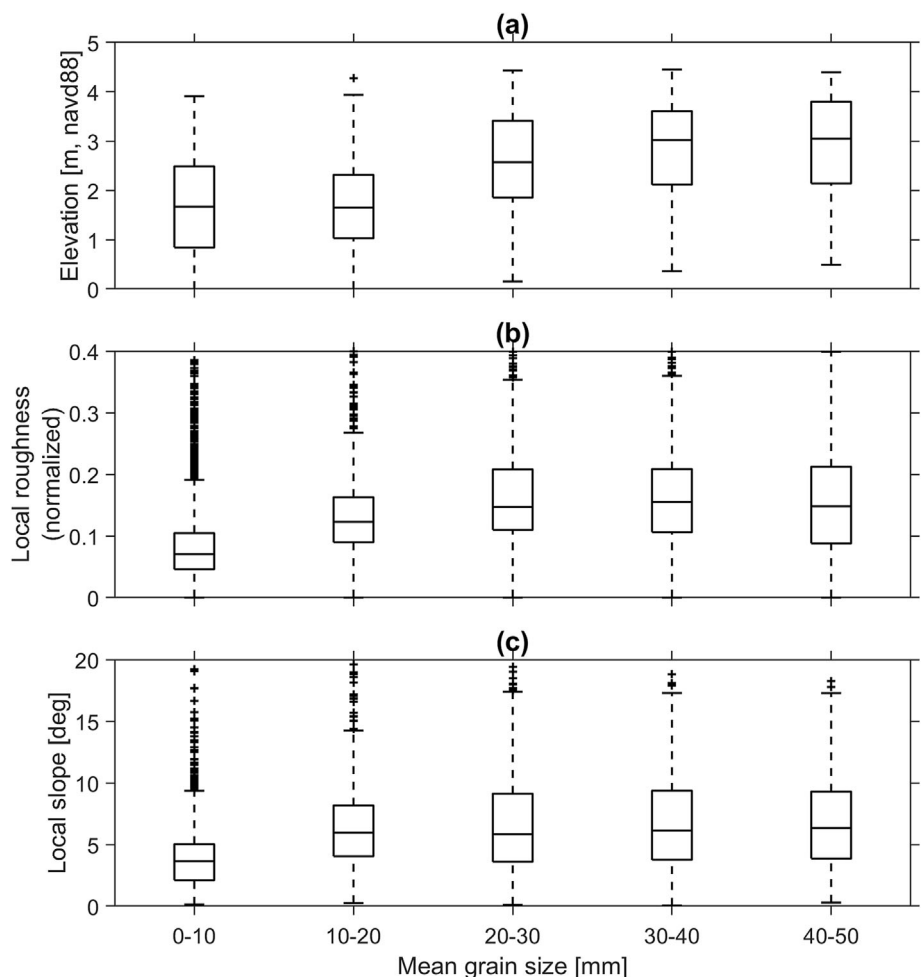


FIGURE 9 Box plots comparing mean mobile digital grain size (MDGS) grain sizes versus (a) elevation, (b) local roughness and (c) local slope. Parameters were obtained from rasters (0.2 m ground sampling distance) generated with the LiDAR surveys. The horizontal line in the box, the bottom and top box edges represent the median, and 25th and 75th percentile, respectively. '+' markers indicate outliers.



occurred (Figure 10b,c,e,f). However, portions of Profile-1 (15–25 m cross shore) and Profile-2 (20–25 m cross-shore position) had similar 4–6° slopes but variable grain sizes of 25 and <10 mm, respectively (grey shaded zones, Figure 10e,f). Furthermore, Profile-2 and Profile-3 had similar MSL-MHW slopes (3.3°) but different MSL-MHW grain sizes of around 4 and 27 mm, respectively (Figure 10c,d).

4 | DISCUSSION

4.1 | MDGS capability

This study evaluated the MDGS at a mixed sediment beach consisting of sand and gravels. The MDGS results were generally consistent with

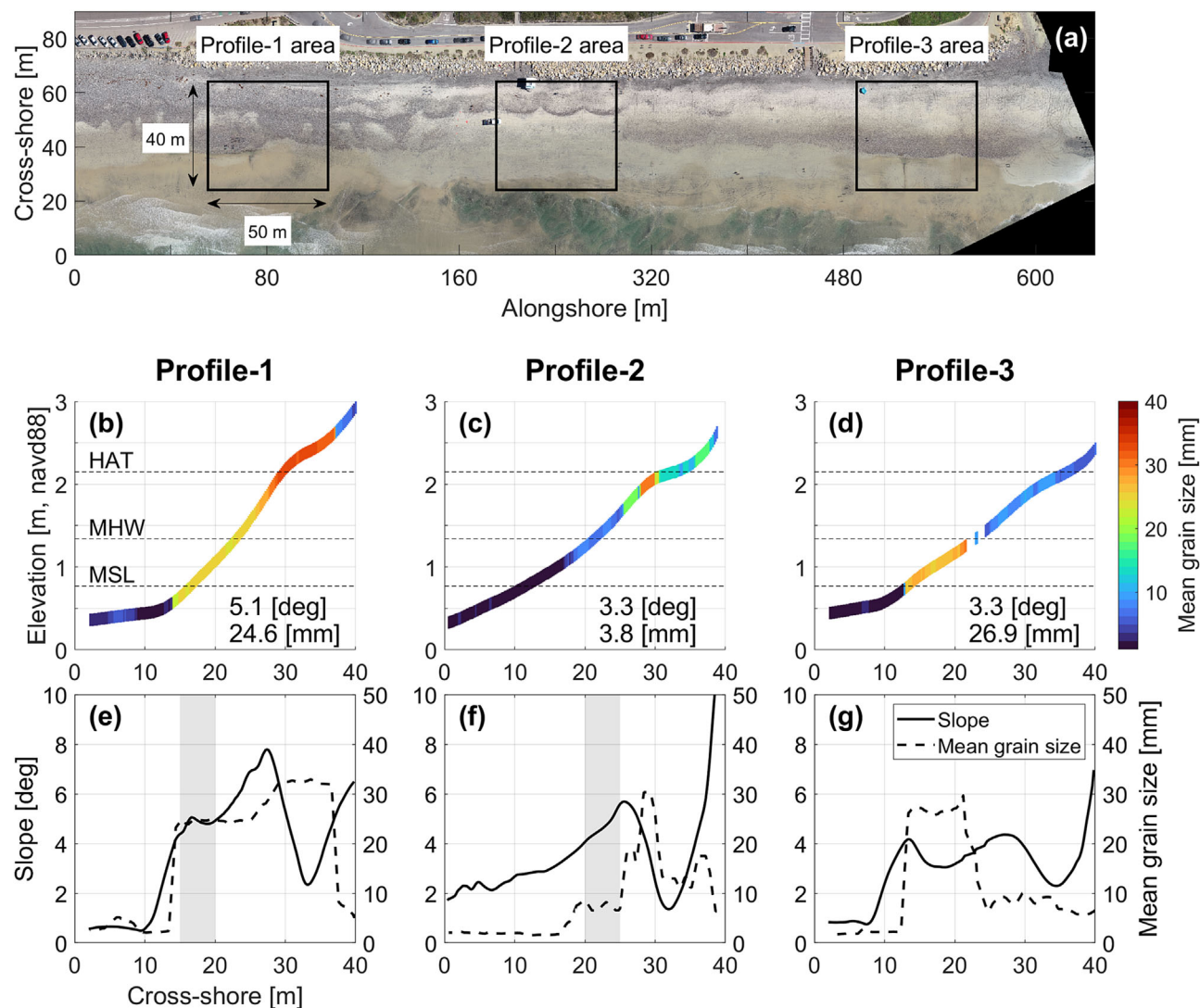


FIGURE 10 (a) Orthomosaic image from the March 2022 all-terrain vehicle (UAV) survey and profile areas. (b–d) March 2022 cross-shore LiDAR profiles and mobile digital grain size (MDGS) surface grain size (colours, shown in Figure 6b), and (e–g) corresponding slope and mean grain size versus cross-shore positions. Profiles and (surface) grain size in (b–d) were generated using a moving average window of 5 m (cross-shore) by 50 m (alongshore). The numbers in the right bottom corners of (b–d) show the mean beach slope and mean surface grain sizes between mean sea level and mean high water. Grey shades in (e–f) show the cross-shore positions of Profile-1 and Profile-2 with similar slope but different grain sizes. Smoothing (with a 1-m cross-shore moving average window) was applied to slope and mean grain size plots in (e–g). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

the manual grain size estimations ($r^2 = 0.74$ and $RMSE = 5.9$ mm, Figure 4) and the UAV orthomosaic imagery (Figure 6); however, errors were relatively large for sand and sand–gravel mix surfaces (Figure 4). Further research is needed to examine the use of MDGS for estimating sand grain sizes, such as comparing MDGS with manually measured sand size distributions (instead of using a single estimated mean sand grain size). The seasonal behaviours in spatial fine/coarse sediment exposure (Figure 7), the general seasonal fluctuations both in nearshore wave height and spatially averaged beach grain sizes (Figure 8), and the similar increasing trends both in grain size and beach elevation/local slopes (Figure 9a,c) are largely consistent with previous morphological and sedimentological studies of Torrey Pines State Beach (Matsumoto et al., 2020a, 2020b), supporting MDGS application for morpho-sedimentary mixed sediment beach studies. The MDGS also allowed a large number of grain size estimations within a single study site (e.g., Figure 6), highlighting the possibility of spatially and temporarily high-resolution gravel grain size estimations over large spatial areas.

The MDGS results showed the complex relationship locally between beach slope and surface grain sizes (Figure 10) particularly for larger sediments (e.g., >20 mm, Figure 9c). This is consistent with the studies by Bujan et al. (2019) and Woodruff et al. (2021) who found complex beach slope–grain size relationships for beaches with variable sediments ranging from sand to boulders. Woodruff et al. (2021) also found little trend in the slope–median grain size relationship for bimodal (e.g., mixed sand–gravel) beaches. The MDGS provides a new method to investigate complex beach grain size–slope relationship and related coastal hazards at mixed sediment beaches. For example, better understanding of grain size–slope relationships for mixed sediment beaches are needed to improve estimates of wave runup and coastal flooding (e.g., Blenkinsopp et al., 2022; Fiedler et al., 2020).

The MDGS has both advantages and limitations compared with existing mobile grain size estimation methods. For example, aerial-based methods (e.g., airborne-LiDAR and UAV-photogrammetry) generally cover large areas with few spatial gaps, although these methods

are often only capable of resolving coarse sediments (e.g., gravels) owing to relatively low-resolution data (e.g., Carboneau et al., 2004, 2018; Gómez et al., 2022; Woodget & Austrums, 2017). The MDGS can map both fine and coarse sediment surfaces but often contains spatial gaps (e.g., Figure 6). Matsumoto and Young (2018) used mobile terrestrial LiDAR offering considerably higher resolution data (10^2 – 10^3 points per square meter) than the MDGS, although their method only allows binary mapping (either 'sand' or 'cobble'). UAV surveys are often hindered by high beach crowds or adverse weather (e.g., strong wind), while the MDGS is ground-based and more operationally robust.

The MDGS can aid existing approaches to collect beach grain size information and build grain size databases. McFall et al. (2020) initiated a community engagement project using the updated DGS by Buscombe (2020) applied to crowd-sourced digital photographs (SandSnap; <https://sandsnap-erdcchl.hub.arcgis.com/>), that currently collects digital photographs globally. The MDGS is fully automated including photo-collection with no manual adjustment and calibration and therefore can broadly collect grain size information from mixed sediment beaches by attaching the camera system to ground-based vehicles that regularly operate on beaches (e.g., lifeguard vehicles).

4.2 | MDGS improvements

Further improvement of the MDGS is possible. For example, the MDGS estimation errors for sand-only and sand-gravel mix surface sediments were relatively large (Figure 4). This is partly likely because the relatively large distance between the camera and ground prevented resolving finer sediments in the digital photographs. To improve grain size estimation skill of sand-only and sand-gravel mix photographs, adding more camera(s) closer to the ground (e.g., a similar configuration to Prodger et al., 2017) should provide improved resolution of finer sediments.

Spatial data gaps existed (Figure 6b) primarily because many results were omitted owing to poor quality of digital photographs (i.e., blurry photographs), photographs with limited number of grains and photographs with meso-scale topographic irregularities (e.g., beach depressions, Figure 2). The results also showed that photographs with 'rough sand surface' and 'non-sediment' can increase MDGS errors (Figure 5). Camera to ground distance and resultant GSD estimation errors may exist owing to LiDAR and GPS accuracy. Previous studies also suggest that variability in lighting (and consequent shadows) can cause inaccurate DGS estimations (e.g., Buscombe et al., 2010; Warrick et al., 2009). To address these issues, future data collection could (1) use a more frequent photo-collection system (currently every 3 s) to reduce the spatial data gaps, (2) implement an automated algorithm to detect and remove photographs with 'rough sand surface' and 'non-sediment', (3) use a more accurate GSD estimation system (e.g., range finder) and (4) use a light controlling system that reduces lighting variability.

5 | CONCLUSIONS

To better understand the behaviours of beaches with complex and variable grain size composition, high spatial and temporal grain size

estimations are needed, as exemplified by previous and present investigations of grain size versus beach slope relationships. This knowledge is critical for coastal hazard studies at mixed sediment beaches and projects using coarse sediments to increase beach stability. However, previous beach grain size studies are often spatially and temporally sparse. This study developed and tested an automated MDGS analysis method using an existing (nonmobile) DGS by Buscombe et al. (2010) at a mixed sediment beach consisting of sand and gravels. Using digital photographs collected approximately monthly for 15 months, the MDGS agreed with manual grain size estimations ($n = 962$, $r^2 = 0.74$ and $RMSE = 5.9$ mm) and classified UAV orthomosaic imagery. Errors were relatively large for sand-only and sand-gravel mix surfaces, and further studies are needed because of uncertainties in estimating sand grain sizes using MDGS. MDGS seasonal trends and general increasing trends both in MDGS grain size, nearshore wave estimates, and beach elevation/local slope highlight the potential of MDGS to conduct spatially and temporarily high-resolution gravel grain size estimations and improved morpho-sedimentary studies of mixed sediment beaches. Future improvements could include using multiple cameras, applying shorter time intervals between photo taking, and developing improved methods to automatically remove erroneous photographs, estimate GSD more accurately and reduce lighting variability. The successful application of MDGS for conducting high temporal and spatial resolution studies will advance the understanding of mixed sediment beaches.

AUTHOR CONTRIBUTIONS

Hironori Matsumoto conceived the study and conducted investigations including data collection and data analysis. Adam P. Young supported data collection and funding acquisition. Hironori Matsumoto prepared the initial manuscript and further revised the manuscript based on input from Adam P. Young.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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